Chapter 13: Dental Biomaterials

Comprehensive NBDHE Board Exam Review — Questions, Answers & Rationales (100 Questions)

Question 1

Which mixing method is used for some VPS impression materials?

a. Capsule in triturator
b. Mixing nozzle
c. Hand-mix
d. Agitation using vibration
e. Spatula and a mixing bowl

Correct Answer: A — Capsule in triturator

Clinical Rationale: a. Powder and liquids are pre-capsulated for GI/RMGI materials, dental amalgam, and some temporary cements. b. Two liquids are mixed through a nozzle for impression materials and for some polymer-based cements c. Powder and liquid are mixed together on a mixing pad for some cements d. Dental materials are not mixed together simply by vibration e. Powder and liquid are mixed together in a mixing bowl for plaster, stone, die stone, and alginate impression material

Question 2

What is the initial setting time?

a. Elapsed time from the start of mixing to complete setting
b. Elapsed time from the start of the working interval until the end of the setting interval
c. The setting interval
d. Time at which sufficient reaction has occurred to cause the materials to be resistant to further manipulation
e. The time available by the operator to place a dental material

Correct Answer: A — Elapsed time from the start of mixing to complete setting

Clinical Rationale: a. This is the sum of the mixing and setting intervals, and not the initial setting time b. This is the sum of the working and setting intervals, and not the initial setting time c. This is the time from the start to the end of the setting process, and not the initial setting time d. Time at which sufficient reaction has occurred to cause the materials to be resistant to further manipulation e. This is the working interval, and not the initial setting time

Question 3

Which one is not a type of chemical bonding?

a. Metallic
b. Micromechanical
c. Hydrogen
d. Ionic
e. Covalent

Correct Answer: A — Metallic

Clinical Rationale: a. Positively charged metal ions are surrounded by cloud of electrons, this is a type of chemical bonding b. Micromechanical bonding is a type of mechanical bonding. For example, enamel etching dissolves minor amounts of the surface hydroxyapatite crystals to provide microretention for the dental adhesive c. Attraction between a hydrogen atom in one molecule and a small electronegative atom in another molecule, this is a type of chemical bonding d. Formed by complete transfer of electrons thus resulting in a structure that is strongly bonded where electrons are shared between specific ions, this is a type of chemical bonding e. Formed by sharing of electrons between atoms thus the formed bond is directional, this is a type of chemical bonding

Question 4

Which material does NOT set by polymerization?

a. RMGI cement
b. Pit-and-fissure sealant
c. Wax
d. Flowable dental composite
e. VPS

Correct Answer: A — RMGI cement

Clinical Rationale: a. RMGI cement contains monomers that undergo polymerization during setting b. Pit-and-fissure sealant contains resin monomers that undergo polymerization during setting c. Wax does not contain monomers and does undergo a setting reaction d. Flowable dental composite contains resin monomers that undergo polymerization during setting e. VPS sets by polymerization to become a polymeric flexible material

Question 5

Which material involves imbibition problems?

a. Amalgam
b. Alginate impression material
c. Flowable composite
d. Zirconia
e. Polyether impression material

Correct Answer: A — Amalgam

Clinical Rationale: a. Amalgam does not contain water b. Alginate is a hydrogel that contains almost 85 percent water and quickly gains (imbibition) or loses water depending on the environment in which it is c. Flowable composite absorbs minimal water after polymerization, but does not undergo problematic imbibition d. Zirconia is a high-strength ceramics that does not gain or lose water e. Polyether is intrinsically hydrophilic; even if it contains some water in the composition, does not undergo problematic imbibition

Question 6

Which restoration is a good thermal conductor?

a. Composite veneer
b. PFM crown
c. Glass ionomer cement
d. CAD/CAM ceramic restoration
e. Zirconia inlay

Correct Answer: A — Composite veneer

Clinical Rationale: a. Composite veneers are insulating b. PFM crowns conduct heat relatively easily because they have a metal core c. Glass ionomer cements involve glass and polymeric phases which are insulating d. CAD-CAM ceramic restorations are very poor thermal conductors e. Zirconia restorations are very poor thermal conductors

Question 7

Which material may undergo crevice corrosion under plaque?

a. Glass-matrix ceramics
b. VPS impression material
c. Gold restoration
d. Polycarboxylate cement
e. Zirconia

Correct Answer: A — Glass-matrix ceramics

Clinical Rationale: a. Glass-matrix ceramics does not contain metal, does not undergo crevice corrosion b. VPS impression material is not a metal c. Gold restorations undergo crevice corrosion under plaque films d. Polycarboxylate cement is not a metal, does not undergo crevice corrosion e. Zirconia is a ceramic, not metal, and does not undergo crevice corrosion

Question 8

Which term is not related to color?

a. BPA
b. Munsel
c. Metamerism
d. L∗a∗b∗
e. VITA

Correct Answer: A — BPA

Clinical Rationale: a. BPA or bisphenol-A is not related to color. It a residue found in pit-and-fissure sealants and composite resins b. Munsel is a color scale c. Metamerism occurs when two colors with different spectral distributions look the same under certain conditions d. L∗a∗b∗ refers to the three components (L*, a*, b*) of color in the CIE system e. VITA is a shade guide using for matching color developed by the VITA company

Question 9

Which one of the following is not associated with electrochemical corrosion?

a. Crevices
b. Plaque
c. Saliva
d. Passivation
e. Monomer

Correct Answer: A — Crevices

Clinical Rationale: a. Crevice corrosion is electrochemical corrosion in the crack under plaque, between a restoration and the tooth structure b. Plaque is involved in crevice corrosion c. Saliva serves as the electrolyte in electrochemical corrosion d. Passivation is when the product of corrosion forms a protective layer on the surface preventing the underlying material from further chemical interaction e. Monomers are not associated with electrochemical corrosion. Monomers form a matrix that is converted to polymer on setting, as in composite resins.

Question 10

What is a material’s strength?

a. Maximum deformation before failure
b. Maximum elastic behavior before failure
c. Toughness
d. Highest stress supported before failure
e. Resilience

Correct Answer: A — Maximum deformation before failure

Clinical Rationale: a. Maximum deformation before failure is the elongation b. Maximum elastic behavior before failure is the total elastic strain c. Toughness is the energy absorbed up to the point of failure d. A material’s strength is the highest stress supported before failure e. Resilience is the total elastic energy absorbed up to the elastic limit

Question 11

Engineering stress is computed as:

a. Modulus multiplied by the plastic deformation
b. Applied load divided by original cross-sectional area
c. Difference between the total strain and plastic strain
d. Deformation divided by the original length
e. Deformation divided by the volume of the object

Correct Answer: A — Modulus multiplied by the plastic deformation

Clinical Rationale: a. Modulus multiplied by the plastic deformation does not make sense b. Engineering stress is the applied load divided by the original cross-sectional area of the object being loaded c. The difference between the total strain and the plastic strain is the elastic strain d. The deformation divided by the original length is the total strain e. Deformation divided by the volume of the object does not make sense

Question 12

Engineering strain is computed as:

a. Load at yield divided by elastic deformation
b. Applied load divided by the total elastic deformation
c. Change between the initial and final strain divided by the time
d. Area under the stress-strain curve
e. Deformation divided by the original length of the object

Correct Answer: A — Load at yield divided by elastic deformation

Clinical Rationale: a. Load at yield divided by elastic deformation does not make sense b. Applied load divided by the total elastic deformation does not make sense c. Change between the initial and final strain divided by the time does not make sense d. Area under the stress-strain curve does not make sense e. Engineering strain is the deformation divided by the original length of the object, or the change in length per unit of length of the object

Question 13

What is another name for a material’s elastic modulus?

a. Fracture strength
b. Fatigue resistance
c. Stiffness
d. Brittleness
e. Toughness

Correct Answer: D — Brittleness

Clinical Rationale: a. Fracture strength is the maximum stress supported prior to failure b. Fatigue resistance is the ability to resist many cycles of stress under all loading conditions c. Elastic modulus or stiffness of a material measures the stress required to produce strain (and is the slope of the stress-strain) d. Brittleness is defined as very limited ability to undergo plastic deformation prior to fracture e. Toughness is the area under the stress-strain curve up to the breaking point

Question 14

What is the definition of a noncarious cervical lesion?

a. Fracture of amalgam margins resulting from creep and occlusal loading
b. Wear of buccal surfaces of posterior teeth
c. Fractures oriented parallel to the long dimension of a tooth around the lingual surface
d. Saucer or notch-shaped cervical tooth loss due to tooth flexure and other factors
e. Loss of a posterior tooth cusp due to cracks coming from an amalgam restoration

Correct Answer: A — Fracture of amalgam margins resulting from creep and occlusal loading

Clinical Rationale: a. Fracture of amalgam margins resulting from creep and occlusal loading is not related to non-carious lesions in the cervical area of teeth b. Noncarious cervical lesions are multifactorial and involve posterior and anterior teeth c. Noncarious cervical lesions are common on the buccal surfaces d. Noncarious cervical lesions occur as saucer or notch-shaped cervical tooth loss due to tooth flexure and other factors e. Loss of a posterior tooth cusp due to cracks coming from an amalgam restoration is not related to noncarious cervical lesions

Question 15

What property is measured when a pyramid-shaped indenter is applied to the surface of a dental material?

a. Toughness
b. Resilience
c. Ultimate strength
d. Hardness
e. Fatigue

Correct Answer: A — Toughness

Clinical Rationale: a. Toughness is the area under the stress-strain curve which estimates the energy to failure b. Resilience is the area under the stress-strain curve up to the elastic limit c. Ultimate strength is the highest stress supported before failure d. Hardness is a measure of materials resistance to localized plastic deformation caused by indentation. Vickers and Knoop hardness are measured with a pyramid-shape indenter e. Fatigue is the accumulation of plastic strain during repeated load cycling

Question 16

What is the approximate Knoop hardness value for enamel?

a. 460
b. 340
c. 60
d. 100
e. 370

Correct Answer: A — 460

Clinical Rationale: a. 460 is the approximate Knoop hardness value for feldspathic ceramic b. 340 is the approximate Knoop hardness value for enamel c. 60 is the approximate Knoop hardness value for dentin d. 100 is withing the range of the approximate Knoop hardness value for composite resin e. 370 is approximate Knoop hardness value for gold (type IV)

Question 17

Which method of amalgam mixing minimizes mercury vapor escape?

a. Covered mixing arm on the triturator
b. Mortar and pestle
c. Friction-fit capsule
d. Precapsulated alloy and mercury
e. More rapid trituration

Correct Answer: A — Covered mixing arm on the triturator

Clinical Rationale: a. Covered mixing arm on the triturator prevents the formation of aerosols if leakage occurs during mixing but does not prevent mercury vapor escape b. Mortar and pestle mixing is an antiquated mixing approach and is no longer recommended because it is inefficient and permits mercury vapor release c. Friction-fit capsules are very rarely used and tend to leak during the trituration process d. Precapsulated alloy and mercury minimize mercury release e. More rapid trituration increases the likelihood of mercury release

Question 18

What agency regulates exposure to mercury vapor in air during a 40-hour work week?

a. EPA
b. CDC
c. ADA
d. OSHA
e. FDA

Correct Answer: A — EPA

Clinical Rationale: a. EPA (Environmental Protection Agency) regulates air and water pollution b. CDC (Centers for Disease Control and Prevention) manage public health challenges c. ADA (American Dental Association) develops dental product standards d. OSHA (Occupational Safety and Health Administration) regulates worker exposure limits e. FDA (Food and Drug Administration) protects public health by ensuring the safety, efficacy, and security of human and veterinary drugs, biological products, and medical devices

Question 19

What is a major problem associated with polishing amalgam restorations?

a. Burnishing of surface corrosion products into the amalgam
b. Production of grayish color
c. Smearing amalgam onto enamel to cause staining
d. Creation of marginal ditching
e. Localized surface melting of the amalgam with mercury smearing

Correct Answer: A — Burnishing of surface corrosion products into the amalgam

Clinical Rationale: a. Surface corrosion products including tarnish are easily removed by polishing and do not become embedded into the amalgam b. Polishing produces a shiny lustrous surface that is mirror-like c. Amalgam smear layers are easily removed and do not produce enamel staining d. Polishing will not produce marginal ditching e. Polishing produces heat that almost invariably causes localized melting of the amalgam matrix of Ag-Hg at the surface

Question 20

Which guidelines advise on dental amalgam handling in the office and include polishing?

a. EPA water quality standards
b. EPA air quality standards
c. ADA standards for recapture devices
d. Best management practices
e. Recycling of amalgam capsules

Correct Answer: A — EPA water quality standards

Clinical Rationale: a. The EPA is not directly involved with best management practices b. The EPA is not directly involved with best management practices c. The ADA standards are only part of best management practices d. Best management practices advise on all details of handling dental amalgam and mercury e. Used amalgam capsules contain residual traces of mercury and mixed amalgam and should be carefully stored and recycled, but are only part of the overall best management practices

Question 21

What is the principal reason for the shift away from dental amalgam use?

a. Composite cost is much lower
b. Patients prefer composite restorations
c. Patients desire all-ceramic restorations
d. Environmental concerns, Minamata agreement commits to reduce worldwide Hg use
e. Improved dental adhesive techniques

Correct Answer: A — Composite cost is much lower

Clinical Rationale: a. The cost of a dental procedure with composite resin is higher than a dental amalgam procedure b. While patients prefer tooth-colored restorations, amalgam restorations may be indicated in specific clinical situations c. Patients like all-ceramic restorations, but they are much more costly d. Environmental concerns, Minamata agreement to reduce all Hg use worldwide e. While dental adhesive techniques continue to improve, they are not directly related to decreased amalgam utilization

Question 22

Which direct restorative material may be substituted for dental amalgam in posterior restoration applications on the occlusal surface of permanent teeth?

a. Glass ionomer
b. Gold crown
c. Composite resin
d. Ceramic restoration
e. ART restoration

Correct Answer: A — Glass ionomer

Clinical Rationale: a. Glass ionomer restorations do not have sufficient fracture resistance to be used in high stress and high wear b. Gold crowns are a good substitute for dental amalgam, but they are significantly more costly and involve a less conservative preparation c. Composite resins have excellent strength and wear resistance, are esthetic, can be easily repaired, and are relatively low-cost d. Ceramic restorations, including lithium disilicate and zirconia restorations, are very popular for posterior use but much more costly and involve a less conservative preparation e. ART restorations are long-term temporary restorations to treat patients living in the underserved regions of the world

Question 23

What is the reason an older class V amalgam may have surfaces slightly elevated above surrounding tooth structure?

a. Amalgams are often poorly placed in class V sites
b. Excessive corrosion causes amalgam creep on facial surfaces
c. Secondary caries lesions may help extrude the amalgam surface
d. Amalgams expand very slowly and without abrasion become extruded
e. Reaction of F ions with amalgam produces excessive surface expansion

Correct Answer: A — Amalgams are often poorly placed in class V sites

Clinical Rationale: a. Even if amalgams are poorly placed, Class V sites are not unusually difficult to restore to obtain good margins b. Corrosion occurs slowly but is not related with amalgam surfaces becoming slightly elevated c. Secondary caries lesions do not contribute to amalgam dimensional changes d. Amalgams very slowly expand and without abrasion become slightly extruded e. Amalgam does not react with F ions

Question 24

Which composite has the smallest average filler particle?

a. Hybrid
b. Nanofilled
c. Microfilled
d. Nanohybrid
e. Monochromatic

Correct Answer: A — Hybrid

Clinical Rationale: a. Hybrid composites contain fillers measuring 0.04 µm to 5.0 µm; the average particle size of current hybrid materials ≤ 1 µm, may include pre-polymerized particle clusters b. Nanofilled composites contain fillers measuring from 0.001 µm to 0.1 µm with multiple pre-polymerized particle clusters c. Current microfilled composites contains fillers measuring 0.2 µm to 0.4 µm with pre-polymerized particle clusters d. Nanohybrid composites are hybrid composites e. The term monochromatic is related with color, not with filler size composites are one-shaded composite materials

Question 25

How many shades does a monochromatic composite resin kit carry?

a. 5
b. 3
c. 1
d. 2
e. Depends on the commercial brand

Correct Answer: A — 5

Clinical Rationale: a. Monochromatic composites carry only one shade b. Monochromatic composites carry only one shade c. Monochromatic composites carry one shade d. Monochromatic composites carry only one shade e. Monochromatic composites carry only one shade

Question 26

Which one of the following is true about pit-and-fissure sealants?

a. Sealants are effective if retained in pits and fissures
b. Tinted sealants are less effective than colorless ones
c. Sealant inspection is unnecessary after 2 years
d. Partial sealing of pits and fissures is better than no sealing at all
e. Uncured light-cured sealant along air-exposed surfaces will finish curing in 24 hours

Correct Answer: A — Sealants are effective if retained in pits and fissures

Clinical Rationale: a. Sealants are very effective to prevent caries lesions in the fissure system of posterior teeth if they are fully retained in the fissure b. Tinted sealants are as effective as colorless ones c. Sealants should be re-inspected every 1 to 2 years to ensure that there are no failures that might increase caries risk d. Partial sealing of pits and fissures may encourage development of caries lesions because sealants only fill the uppermost portions of fissures so bacterial leakage could occur underneath e. Uncured sealant along air-exposed surfaces is oxygen-inhibited material that will not cure on its own and should be wiped away at the end of the procedure 24 hours

Question 27

Which of the following is FALSE about composite resins?

a. Smaller particles decrease wear resistance
b. Rough composite surfaces may retain plaque biofilm
c. Wear resistance increases with increased filler content
d. Closer interparticle spacing of fillers increases wear resistance
e. Composite resins are thermal and electrical insulators

Correct Answer: A — Smaller particles decrease wear resistance

Clinical Rationale: a. Smaller particles decrease wear resistance b. Rough composite surfaces may retain plaque biofilm c. Wear resistance increases with increased filler content d. Closer interparticle spacing of fillers increases wear resistance e. Composite resins are thermal and electrical insulators a. Coronal margins of Class V composite restorations are the strongest due to the presence of etched enamel margin b. Current composite resins have small particle size and thus surfaces remain clinically smooth c. Staining of the cervical margin occurs over time because there is no enamel present to provide reliable adhesion d. Current composite resins are formulated to resist discoloration e. Class V composite restorations are strong enough to resist fracture during tooth flexure

Question 28

Which of the following is most likely observed with class V composite restorations over time?

a. Ditching of the coronal margin
b. Surface roughness over a few years
c. Staining of the cervical margin
d. Resin discoloration over a few years
e. Fracture of the restoration

Correct Answer: A — Ditching of the coronal margin

Clinical Rationale: a. Current composite resins are radiopaque b. Zirconia crowns are radiopaque c. Dental amalgam is very radiopaque d. Most dental adhesives do not contain radiopaque fillers e. Glass ionomer restorations are radiopaque because they contain alumino-silicate glass

Question 29

Which one of the following is likely to be deceptive in appearance in a radiograph?

a. Dental composite restoration
b. Zirconia crown
c. Amalgam restoration
d. Dental adhesive
e. Glass ionomer restoration

Correct Answer: A — Dental composite restoration

Clinical Rationale: a. The infiltrant stops progression of initial caries lesions, but is not intended to repair damage of secondary caries lesions b. The infiltrant is for impregnating enamel white spot lesions, does not work as a pit-and-fissure sealant c. The infiltrant is not intended as a repair material d. The infiltrant fills in the spaces underneath the surface of enamel white spot lesions (non-cavitated) to stop progression of caries lesion e. The infiltrant is not intended to seal over SDF treated tooth structure

Question 30

Which one of the following is the primary use for resin infiltrant?

a. Repair of secondary caries at the margin of a composite
b. An alternative pit-and-fissure sealant
c. Repair of deep proximal restoration leakage
d. Infiltration of noncavitated caries lesion
e. Sealer over lesions treated with sodium diamine fluoride

Correct Answer: A — Repair of secondary caries at the margin of a composite

Clinical Rationale: a. 35% phosphoric acid is used to etch enamel prior to the application of a dental adhesive on enamel. It is unable to create microporosities of the hypermineralized surface layer of enamel white spot lesions b. 5% hydrofluoric acid is used to create microretention on the intaglio of glass-matrix ceramic restorations c. 5% phosphoric acid is not currently used for enamel adhesive procedures d. 15% hydrochloric acid is the recommended buffered acid gel capable of creating microporosities of the hypermineralized surface layer of enamel white spot lesions e. 15% phosphoric acid is not currently used for enamel adhesive procedures

Question 31

What is the etching gel used for resin infiltration of enamel white spot lesions?

a. 35% phosphoric acid
b. 5% hydrofluoric acid
c. 5% phosphoric acid
d. 15% hydrochloric acid
e. 15% phosphoric acid

Correct Answer: A — 35% phosphoric acid

Clinical Rationale: a. Sealant application involves enamel bonding only and routinely relies on phosphoric-acid–solution etching b. As opposed to etch-and-rinse adhesives, self-etch adhesives do not rely on phosphoric acid etching c. All composite resin restorations rely on enamel etching with phosphoric acid d. Etch-and-rinse adhesives always include the phosphoric acid etching step e. Porcelain veneers are bonded to enamel using and etch-and-rinse adhesive

Question 32

Which procedure does NOT require phosphoric acid etching?

a. Dental sealant
b. Self-etch dentin bonding procedure
c. Composite veneer
d. Etch-and-rinse bonding procedure
e. Porcelain veneer

Correct Answer: C — Composite veneer

Clinical Rationale: c. Self-adhesive resin cements and glass ionomer cements bond chemically or via specific self-adhesive monomers without requiring separate phosphoric acid etching.

Question 33

What is required to produce good bond strength to dentin?

a. Drying dentin for 20 seconds
b. Preapplication of chlorhexidine
c. Hybrid layer formation
d. Phosphoric acid etching
e. Longer curing times

Correct Answer: C — Hybrid layer formation

Clinical Rationale: a. Dentin is intrinsically humid. There is no need to dry dentin or leave it visibly moist b. There is no clinical evidence that a chlorhexidine cavity cleanser improves adhesion c. Dentin adhesion depends on the formation of microscopic resin tags within intertubular dentin to produce a hybrid zone or hybrid layer that microscopically intertwines collagen bundles and the adhesive d. Self-etch adhesives also form hybrid layers with dentin e. Longer curing times are not related with the formation of dentin hybrid layers

Question 34

What is required to produce good bond strength to enamel?

a. Leaving enamel visibly moist
b. Longer curing times
c. Hybrid layer formation
d. Phosphoric acid etching
e. Preapplication of primer

Correct Answer: A — Leaving enamel visibly moist

Clinical Rationale: a. Enamel must be dried to provide higher bond strengths b. Curing times longer than those recommended by the respective manufacturer are not related with good bond strength to enamel c. Although a very thin hybrid layer may form within enamel, it does not improve bond strengths d. Phosphoric acid etching is crucial to obtain good bond strengths to enamel e. Pre-application of a primer is not associated with higher bond strengths to enamel

Question 35

How do recent self-etch adhesives work?

a. Wetting agent helps adaptation and replaces need for micromechanical bonding
b. New monomers chemically adhere to calcium in hydroxyapatite
c. Phosphoric acid is mixed with other components of the bonding system
d. Chemical bonding to collagen replaces need for micromechanical bonding
e. Self-etch adhesives bond to dentin prisms

Correct Answer: A — Wetting agent helps adaptation and replaces need for micromechanical bonding

Clinical Rationale: a. Wetting agents are not relevant for the interaction of self-etch adhesives with enamel and dentin b. Recent self-etch adhesives bond chemically to calcium ions in the residual smear layer embedded into the hybrid layer c. Phosphoric acid etching is not used with self-etch adhesives d. There is no chemical bonding to collagen e. Prisms the structural units of enamel

Question 36

How do universal adhesive systems work?

a. Can only be used as ER adhesives
b. Are applied to enamel and dentin in three steps
c. Contain functional monomers that adhere chemically to hydroxyapatite
d. Chemical bonding to collagen replaces need for micromechanical bonding
e. Bond chemically to gold restorations

Correct Answer: A — Can only be used as ER adhesives

Clinical Rationale: a. They can be used as ER or as SE adhesives b. They are applied in 1 step (SE) or in 2 steps (ER) c. They contain functional monomers, such as 10-MDP, that bond chemically to hydroxyapatite d. There is no chemical bonding to collagen e. There is no chemical bonding to gold

Question 37

What product most likely contains hydrophilic (water-miscible) resin monomers?

a. Dentin primer
b. APF
c. Pit-and-fissure sealant
d. Resin surface sealer
e. Composite resin

Correct Answer: A — Dentin primer

Clinical Rationale: a. Dentin primer contains hydrophilic monomers such as HEMA b. APF does not contain resin monomers c. Pit-and-fissure sealant do not contain water-miscible monomers d. Resin surface sealer e. Composite resin is hydrophobic, does not contain hydrophilic monomers

Question 38

Which component of a dental adhesive system is most likely to cause skin sensitization in dental personnel?

a. Ethanol
b. UDMA
c. Eugenol
d. BisGMA
e. HEMA

Correct Answer: A — Ethanol

Clinical Rationale: a. Ethanol is a water-soluble solvent used in dental adhesives, but it does not produce sensitization b. UDMA is a monomer in many composite resins, but it generally does not produce sensitization c. Eugenol is not a component of dental adhesives d. BisGMA is a monomer in many dental adhesives and composite resins, but it does not usually produce sensitization e. HEMA is a common monomer in dental adhesive primers that causes relatively rapid sensitization in some patients and dental personnel

Question 39

What is the shorthand representation for a three-step etch-and-rinse adhesive?

a. E
b. EP + B
c. E + P + B
d. E + PB
e. EPB

Correct Answer: A — E

Clinical Rationale: a. E stands for etching b. EP+B stands for a 2-step self-etch adhesive c. E+P+B stands for a 3-step etch-and-rinse adhesive (Etching+Primer+Bonding) d. E+PB stands for 2-step etch-and rinse adhesive e. EPB stands for a 1-step self-etch adhesive

Question 40

Which material may require surface protection with petroleum jelly during APF application?

a. Amalgam
b. Glass-matrix ceramic
c. Cast gold
d. Zirconia
e. Dental sealant

Correct Answer: A — Amalgam

Clinical Rationale: a. Amalgam does not dissolve in APF b. Glass-matrix ceramic may undergo a slight surface etching when in contact with APF c. Cast gold is not soluble in APF d. Zirconia does not dissolve in APF because it does not contain a glass phase e. Dental sealant does not dissolve in APF because it is made of resin

Question 41

What is the primary purpose of fluoride in APF surface treatments?

a. Precipitate CaF_2 on tooth surface
b. Incorporate F ion into hydroxyapatite to increase acid resistance
c. Kill bacteria in the biofilm
d. Infiltrate enamel white spot lesions
e. Reinforce pit-and-fissure sealant

Correct Answer: A — Precipitate CaF_2 on tooth surface

Clinical Rationale: a. While CaF_2 precipitates can form, they are not protective and are lost quickly from tooth structure b. F ions replace a few hydroxyl ions in hydroxyapatite and increase the acid resistance c. F may kill bacteria in the biofilm, but this is very temporary d. Only resin infiltrants are able to infiltrate enamel white spot lesions e. There is no effect on pit and fissure sealants

Question 42

What are the recommendations for patients after the application of fluoride varnish?

a. Brush as soon as you get home
b. Do not drink and eat for 12 hours
c. Do not brush or floss for the remainder of the day
d. Do not brush or floss for 4 to 6 hours
e. Rinse with a fluoride solution after 1 hour

Correct Answer: A — Brush as soon as you get home

Clinical Rationale: a. It would remove the fluoride varnish from the surface of teeth without allowing enough time for fluoride diffusion into enamel b. There is no need to avoid eating and drinking for 12 hours c. Patients can brush and floss 4-6 hours after the treatment d. 4-6 hours allow sufficient time for fluoride diffusion from varnish into enamel e. There is no need to rinse with fluoride q hour after fluoride varnish application

Question 43

Which constituent may have a palliative action on the dental pulp?

a. HEMA
b. UDMA
c. BisGMA
d. Calcium hydroxide
e. Eugenol

Correct Answer: A — HEMA

Clinical Rationale: a. HEMA is a monomer that is a sensitizer and is irritating to cells if it diffuses to the pulp b. UDMA is a monomer that is irritating to cells if it diffuses to the pulp c. BisGMA is a monomer that is irritating to cells if it diffuses to the pulp d. Calcium hydroxide stimulates the production of tertiary dentin if it diffuses to the dental pulp, but it is not palliative e. Eugenol produces a palliative action if it diffuses to the dental pulp

Question 44

What is the primary component of most composite fillers?

a. Zirconium oxide
b. Aluminum oxide
c. Zinc oxide
d. Calcium oxide
e. Silica (silicon oxide or silicon dioxide)

Correct Answer: A — Zirconium oxide

Clinical Rationale: a. Zirconium oxide (zirconia) is a very strong ceramic material, also used in a few recent composite resins b. Aluminum oxide (alumina) is not used in composite resins c. Zinc oxide is the major reactant and filler in ZOE cement d. Calcium oxide is not used in composite resins e. Silica (silicon oxide or silicon dioxide) is the primary component of fillers in composite resins

Question 45

Which materials bond chemically to calcium in tooth structure without the need of a separate dental adhesive?

a. Polycarboxylates (PC) and GI cements
b. Composite resin and calcium hydroxide liner
c. Self-adhesive resin cements and conventional resin cements
d. Zinc phosphate cement and self-curing denture bases
e. Compomers and ZOE

Correct Answer: A — Polycarboxylates (PC) and GI cements

Clinical Rationale: a. Polycarboxylates (PC) and GI cements bond chemically to calcium in tooth structure as a result of their polycarboxylic acid component b. Composite resin and calcium hydroxide liner do not bond chemically to tooth structure. Recent self-adhesive composite resins have the potential of bonding chemically to dentin c. While self-adhesive resin cements bond chemically to tooth structure, conventional resin cements do not d. Zinc phosphate cement and self-curing denture bases do not bond chemically to tooth structure e. While a few researchers have claimed that compomers have the potential to bond chemically to tooth structure, zinc-oxide eugenol cements do not

Question 46

Which material does not release fluoride?

a. Resin-modified glass ionomer
b. Giomer
c. Provisional composite restoration
d. ART restoration
e. Traditional glass ionomer

Correct Answer: A — Resin-modified glass ionomer

Clinical Rationale: a. Resin-modified glass ionomers release fluoride b. Giomers are composite resins that contain precured GI particles. They release minimal fluoride c. Provisional composite restoration does not release fluoride d. GI materials are used for ART restorations which explains their fluoride release e. Traditional (resin-free) glass ionomer materials release fluoride

Question 47

Fluoride release from glass ionomer follows what pattern?

a. Slow release at first
b. Slow release after 7 to 30 days
c. Some increase after 3 months
d. No decrease after 6 months
e. Some decrease after 12 months

Correct Answer: A — Slow release at first

Clinical Rationale: a. Rapid release at first because of excess fluoride ions in the matrix b. Slow release after 7 to 30 days because of slow diffusion of fluoride ions out of aluminosilicate particles c. Fluoride release only increases for very short periods is the restoration is recharged with fluoride d. Fluoride release decreases gradually for all GI materials e. Fluoride release decreases gradually for all GI materials

Question 48

What is the effectiveness of a recharging glass ionomer restoration?

a. Does not work
b. Increases F release for a few months
c. Increases F release for 1 to 2 years
d. Permanently reestablishes F release from the material
e. Increases F release for a few days

Correct Answer: A — Does not work

Clinical Rationale: a. Recharging does work but is not very effective b. Recharging has only a short-term effect and not for a few months c. Recharging has only a short-term effect and not for a few years d. Recharging is not permanent e. Recharging has only a very short-term effect and lasts a few days

Question 49

Which statement about resin-modified glass ionomer (RMGI) materials is FALSE?

a. Their esthetics are not as good as that of composite resins
b. Their clinical retention is excellent in class V lesions
c. Their solubility is similar to that of composite resins
d. They release more fluoride than conventional GIs
e. They can be finished immediately after insertion and polymerization

Correct Answer: A — Their esthetics are not as good as that of composite resins

Clinical Rationale: a. Composite resins mimic the esthetics of tooth structure better than RMGI materials b. Clinical trials and systematic reviews have concluded that RMGI restorations result in excellent retention in class V lesions c. Composite resins are less porous and less vulnerable to acidic pH, therefore less soluble d. Because of the presence of a resin monomer grafted into the GI matrix, RMGIs release less fluoride than conventional GIs e. As opposed to conventional GIs, RMGIs can be finished immediately after insertion and polymerization

Question 50

Which of the following applications typically involves an ART material?

a. Esthetic reconstruction of anterior
b. Liner under composite restorations
c. Temporary crowns
d. Long-term temporary restorations after removing the bulk of carious dentin manually
e. Pulpotomy of deciduous teeth

Correct Answer: A — Esthetic reconstruction of anterior

Clinical Rationale: a. The ART technique is not indicated for esthetic reconstructions b. The ART technique is not indicated as liner under composite restorations c. The ART technique is not indicated for temporary crowns d. Glass ionomer is used for ART procedures in underserved regions of the world as long-term temporary restorations for caries control to gain time until patients can get more advanced dental care e. The ART technique is not indicated for pulpotomy of deciduous teeth

Question 51

Which are the important properties of root canal sealers?

a. Insolubility
b. Biocompatibility
c. Radiopacity
d. Tensile strength
e. All of the above

Correct Answer: A — Insolubility

Clinical Rationale: a. Root canal sealers must be insoluble to provide a good seal inside the root canal b. Root canal sealers should be biocompatible to not cause any undesirable response c. Root canal sealers must be radiopaque to be easily observed in radiographs d. Root canal sealers must have good physical properties to resist tensile stresses inside the root canal e. All options are correct

Question 52

Silver diamine fluoride is indicated for

a. active caries lesions of permanent teeth
b. cavitated caries lesions in patients presenting with behavioral or medical management challenges
c. active caries lesions of deciduous teeth
d. caries lesions not in contact with the pulp tissue
e. all of the above

Correct Answer: A — active caries lesions of permanent teeth

Clinical Rationale: a. Active caries lesions of deciduous and permanent teeth to prevent further progression of the lesion b. Extractions may be contra-indicated in patients with behavioral or medical management challenges. SDF can be applied in those cavitated caries lesions c. Active caries lesions of deciduous and permanent teeth to prevent further progression of the lesion d. Direct SDF application on the pulp causes necrosis. SDF is indicated if the caries lesion is not in contact with the pulp tissue f. All options are correct

Question 53

Carbamide peroxide gel in a custom-made tray is used for at-home tooth bleaching prescribed by a dental professional. Which concentration is recommended by the American Dental Association?

a. 10% to 15%
b. 3.4%
c. 10%
d. 5%
e. Up to 45%

Correct Answer: A — 10% to 15%

Clinical Rationale: a. 15% to 20% concentrations cause more frequent tooth sensitivity than 10% b. 3.4% is the concentration of hydrogen peroxide equivalent to 10% carbamide peroxide c. The ADA does not recommend concentrations >10% carbamide peroxide. Clinical evidence supports the use of 10% carbamide peroxide, as higher concentrations cause more tooth sensitivity but do not improve the final outcome d. 5% is only available for hydrogen peroxide e. While concentrations of up to 45% carbamide peroxide are currently available, they can burn the soft tissues and cause tooth sensitivity

Question 54

Which statement about at-home tooth bleaching with 10% carbamide is FALSE?

a. Cost is lower than that of in-office bleaching with hydrogen peroxide
b. Causes more tooth sensitivity than in-office bleaching with hydrogen peroxide
c. Bleaching effectiveness depends on the baseline tooth color
d. Is more effective than OTC adhesive strips impregnated with hydrogen peroxide
e. Requires 2 to 4 weeks of daily application in most cases

Correct Answer: A — Cost is lower than that of in-office bleaching with hydrogen peroxide

Clinical Rationale: a. Cost at-home tooth bleaching with 10% carbamide is lower than that of in-office bleaching with hydrogen peroxide because the treatment is carried out at home by the patient b. At-home tooth bleaching with 10% carbamide causes less tooth sensitivity than in-office bleaching with hydrogen peroxide c. Shades A and B in the VITA shade guide are easier to bleach than shades C and D in the VITA shade guide d. Clinical studies have reported that at-home tooth bleaching with 10% carbamide is more effective than OTC adhesive strips impregnated with hydrogen peroxide e. 2-4 weeks of daily application is effective for most patients, unless their tooth shade is C or D in the VITA shade guide

Question 55

Which one of the following materials is a potent local anticaries agent?

a. Calcium hydroxide
b. Giomer
c. Pit-and-fissure sealant
d. Silver diamine fluoride
e. Zirconium dioxide

Correct Answer: A — Calcium hydroxide

Clinical Rationale: a. Calcium hydroxide stimulates the production of tertiary dentin if it diffuses to the dental pulp but is not a local anticaries agent b. Giomers are composite resins that contain precured GI particles. They release minimal fluoride, but are not a local anticaries agent c. Pit-and-fissure sealants prevent the development of caries lesions but are not a local anticaries agent d. Silver diamine fluoride is a potent anti-caries agent that arrests caries lesions e. Zirconium dioxide (zirconia) is a high-strength ceramic material with no anticaries effects

Question 56

What is a strategy to restore primary teeth with rampant caries lesions?

a. Extraction of all teeth
b. SDF application and GI or RMGI restoration
c. Carious dentin removal and restoration with flowable composite resin
d. Resin infiltration of the lesions
e. Carious dentin removal and repair with calcium hydroxide

Correct Answer: A — Extraction of all teeth

Clinical Rationale: a. Treatment with SDF avoids a common option of extracting teeth b. Carious dentin removal, application of SDF with a brush, and GI or RMGI restoration c. Classical carious dentin removal and restoration is a lengthy procedure for a pediatric patient d. Resin infiltration is only effective in non-cavitated lesions, such as enamel white spot lesions e. Calcium hydroxide is not strong enough as a restoration

Question 57

Which impression material is a hydrocolloid?

a. Polysulfide
b. Polyether
c. Vinyl polysiloxane
d. Condensation silicone
e. Alginate

Correct Answer: A — Polysulfide

Clinical Rationale: a. Polysulfide contains no water b. Polyether contains no water c. Vinyl polysiloxane contains no water d. Condensation contains no water e. Alginate contains water in its set composition

Question 58

What happens to water phase in a set alginate impression?

a. Stable over time
b. Quickly absorbs additional water from air
c. Quickly contaminated by things in air
d. Tends to move toward the bottom of the impression

Correct Answer: A — Stable over time

Clinical Rationale: a. Water is quickly exchanged with air. It is not stable over time b. Quickly usually loses water to the air c. Alginate is not contaminated by things in the air d. Alginate has a stable water distribution, so water does not tend to pool to the bottom e. Water is quickly lost. Wrap the impression in a wet paper towel while manipulating it and before it is poured with plaster

Question 59

Which elastomer impression material in intrinsically hydrophilic

a. VPS
b. Polyether
c. Polysulfide
d. Condensation silicone
e. None of the above

Correct Answer: A — VPS

Clinical Rationale: a. VPS is hydrophobic, contain surfactants to make surface wetting easier b. Polyether is hydrophilic c. Polysulfide is hydrophobic overall d. Condensation silicone is hydrophobic e. None of the above

Question 60

Which statement about vinyl polysiloxane impression material is TRUE?

a. High shrinkage on setting
b. Dimensionally unstable
c. It is unfilled
d. Unable to reproduce fine details
e. The material of choice for final impressions in prosthodontics

Correct Answer: A — High shrinkage on setting

Clinical Rationale: a. It shrinks very little on setting b. It is dimensionally stable upon setting c. Contains fillers to control any potential shrinkage d. It reproduces very fine details of surfaces and is very accurate e. The material of election for prosthodontics as a result of the precise detailed impressions obtained with this material

Question 61

Which application typically requires the strongest gypsum material?

a. Denture fabrication
b. Study casts
c. Master cast
d. Removable die
e. Orthodontic model

Correct Answer: A — Denture fabrication

Clinical Rationale: a. Denture fabrication requires space filling material for molding but also needs to be easily removed at the end of the process b. Plaster is generally used for study models c. A master cast is the first cast produced for capturing the cavity preparation information and occlusion for the production of indirect restorations d. Dies are duplicates of teeth that can be removed from the working cast to more carefully work with indirect restorations being fit onto them and need to be stronger and wear resistant e. Orthodontic models capture all soft- and hard-tissue information about a patient for reference but not for use in producing indirect restorations

Question 62

What is the chemical composition of set gypsum product?

a. Calcium sulfate trihydrate
b. Calcium sulfate dihydrate
c. Calcium sulfate monohydrate
d. Calcium sulfate hemihydrate
e. Calcium sulfate unhydrated

Correct Answer: A — Calcium sulfate trihydrate

Clinical Rationale: a. Calcium sulfate trihydrate does not exist b. Calcium sulfate dihydrate is the reaction product of water and calcium sulfate hemihydrate c. Calcium sulfate monohydrate does not exist d. Calcium sulfate hemihydrate is gypsum that is only partially reacted with water e. Calcium sulfate unhydrated is not stable at room temperature

Question 63

What is the chemical composition of gypsum powder to form models?

a. Calcium chloride
b. Calcium phosphate
c. Calcium carbonate
d. Calcium fluoride
e. Calcium sulfate hemihydrate

Correct Answer: E — Calcium sulfate hemihydrate

Clinical Rationale: a. Gypsum powder for models (or casts) does not contain calcium chloride b. Gypsum powder for models (or casts) does not contain calcium phosphate c. Gypsum powder for models (or casts) does not contain calcium carbonate d. Gypsum powder for models (or casts) does not contain calcium fluoride e. Gypsum powder for models (or casts) is composed of calcium sulfate hemihydrate

Question 64

What is the major difference between plaster and stone powders?

a. Sterilization technique
b. Powder type
c. Color
d. Powder particle packing
e. Crystal structure

Correct Answer: A — Sterilization technique

Clinical Rationale: a. Neither material is sterilized b. Both materials are calcium sulfate hemihydrate powder c. Very minor colorant is artificially added to distinguish the materials d. Different particle shapes produce different packing efficiencies that affect the amount of excess water required for making a suitable mixture e. The materials are the same crystal structure, although slightly different forms

Question 65

Why does plaster require more water for mixing than diestone?

a. Diestone undergoes a different reaction than stone
b. Diestone powder particles are less porous and imbibe less water
c. Diestone chemical reaction generates more heat and requires more cooling
d. Diestone contains fillers
e. Diestone powder particles pack more efficiently

Correct Answer: A — Diestone undergoes a different reaction than stone

Clinical Rationale: a. All gypsum products set by the same chemical reaction b. Powder particles are not porous c. The chemical reactions for plaster and stone are the same d. There are no fillers added to diestone or plaster e. Different particle shapes produce different packing efficiencies that affect the amount of excess water required for making a suitable mixture

Question 66

What is the major component in most dental waxes?

a. Beeswax
b. Ceresin
c. Carnauba
d. Microcrystalline wax
e. Paraffin

Correct Answer: A — Beeswax

Clinical Rationale: a. Beeswax is not the major wax and is added only to induce tackiness b. Ceresin is a modifier but not the major wax c. wax is a modifier that raises the melting point but not the major wax d. Microcrystalline wax is a modifier but not the major wax e. Paraffin is the major component of most waxes

Question 67

In high-gold casting alloys, which element is primarily responsible for hardness?

a. Ag
b. Zn
c. Au
d. Pd
e. Cu

Correct Answer: A — Ag

Clinical Rationale: a. Ag is added to Au to counteract the more orange effect created by Cu b. Zn is added in small amounts to suppress oxidation during melting and casting c. Au is the dominant element in the composition and imparts good corrosion resistance d. Pd is added in minor amounts to adjust the melting temperature and hardness e. Cu increases the hardness of an otherwise relatively soft composition that is mostly Au

Question 68

What element is responsible for producing the corrosion resistance of stainless steel instruments?

a. C
b. Co
c. Ni
d. Cr
e. Fe

Correct Answer: A — C

Clinical Rationale: a. C is added to Fe to produce steel and improve mechanical properties b. Cobalt is added to steels and other alloys but for other reasons c. Nickel is added to steels and other alloys but for other reasons d. Cr instantaneously reacts with O2 and creates a tight adherent film (chromium oxide) that prevents further corrosion (oxidation) of the alloy surface e. Fe is the major element in stainless steel but is not corrosion-resistant

Question 69

Which application does not use gold alloys?

a. Cast alloy bridge
b. Implants
c. Cast alloy crown
d. Cast post and core
e. Cast onlay

Correct Answer: A — Cast alloy bridge

Clinical Rationale: a. Cast gold alloys can be used for fabricating bridges b. Titanium, rather than gold alloys, is used for most implants because of its superior strength and ability to osseointegrate at its surfaces c. Cast gold alloys can be used for fabricating crowns d. Cast gold alloys can be used for post and cores e. Cast gold alloys are still occasionally used for fabricating onlays

Question 70

Which dental materials cannot cause nickel sensitivity?

a. Stainless steel crowns
b. Dental amalgams
c. Some casting alloys
d. Some orthodontic wires
e. Some partial denture framework alloys

Correct Answer: A — Stainless steel crowns

Clinical Rationale: a. Nickel may be part of the composition of stainless steel crowns b. Dental amalgam does not contain nickel c. Nickel is part of some removable nickel chromium casting alloys d. Nickel may be part of the composition of stainless steel orthodontic wires e. Nickel can still be found in some partial denture framework alloys

Question 71

What is the approximate incidence of nickel sensitivity for men?

a. ~6%
b. ~12%
c. ~18%
d. ~24%
e. ~30%

Correct Answer: A — ~6%

Clinical Rationale: a. The incidence of nickel sensitivity is approximately 6% for men and 11%-24% for women b. The incidence of nickel sensitivity is approximately 6% for men c. The incidence of nickel sensitivity is approximately 6% for men d. The incidence of nickel sensitivity is approximately 6% for men e. The incidence of nickel sensitivity is approximately 6% for men

Question 72

Which event contributes to a relatively high nickel sensitivity level in women?

a. Genetic differences
b. Less protective hair during metal contact
c. Greater contact with stainless steel
d. Greater contact with metal coinage
e. Contact dermatitis from nickel-based jewelry

Correct Answer: A — Genetic differences

Clinical Rationale: a. There are no known genetic predispositions to nickel sensitivity b. Hair does not appear to be a protective factor in limiting contact dermatitis to nickel contact c. All genders appear to be at equal risk to contact with stainless steel in the environment d. All genders appear to be at equal risk to contact with nickel-containing coinage e. Women tend to have much greater nickel-based jewelry contact

Question 73

Which of the following is a practical definition of a ceramic?

a. Glassy or crystalline brittle material
b. Material with a Knoop hardness value of 50
c. Any insulating material
d. Material typically used as an abrasive
e. Composition with primarily metallic and nonmetallic elements

Correct Answer: A — Glassy or crystalline brittle material

Clinical Rationale: a. Not all ceramics contain glass b. Ceramics have a much higher Knoop hardness number c. Many insulating materials are not ceramics d. No all ceramics are abrasive e. Ceramic materials are based on major combinations of metallic and non-metallic elements

Question 74

Which ceramics does NOT include a glass phase?

a. Leucite
b. Lithium disilicate
c. Zirconia-reinforced lithium disilicate
d. Translucent zirconia
e. Feldspathic

Correct Answer: A — Leucite

Clinical Rationale: a. Leucite is a reinforced glass-matrix ceramics b. Lithium disilicate is a reinforced glass-matrix ceramics c. Zirconia-reinforced lithium disilicate is a reinforced glass-matrix ceramics d. Translucent zirconia is a polycrystalline ceramics or oxide ceramics without glass phase e. Feldspathic is a glass-matrix ceramics

Question 75

Which of the following is a high-strength ceramics?

a. Feldspathic
b. Titanium
c. Lithium disilicate
d. Zirconia
e. None of the above

Correct Answer: E — None of the above

Clinical Rationale: a. Feldspathic is a glass-matrix ceramics with flexural strength ≈100 MPa b. Titanium is a metal c. Lithium disilicate is a reinforced glass-matrix ceramics with flexural strength of 400-600 MPa d. Zirconia is a polycrystalline high-strength ceramics with flexural strength ≈1,000 MPa e. 'd' is correct

Question 76

Which statement applies to dental ceramics?

a. Does not cause wear of the opponent teeth
b. Has low coefficient of thermal expansion and contraction
c. High-strength ceramics are more esthetic than low-strength ceramics
d. CAD/CAM design and milling are used only for zirconia
e. Poor electrical and thermal insulation

Correct Answer: A — Does not cause wear of the opponent teeth

Clinical Rationale: a. Dental ceramics may cause wear of opposing enamel b. Dental ceramics have low coefficient of thermal expansion and contraction c. High-strength ceramics are stronger but less esthetic than low-strength ceramics d. CAD-CAM design and milling in Dentistry is used many different applications e. Dental ceramics are excellent electrical and thermal insulators

Question 77

Which one of the following restorations does not contain a ceramic material?

a. PFM crown
b. Titanium implant
c. Porcelain veneer
d. Leucite-reinforced crown
e. PICN inlay

Correct Answer: A — PFM crown

Clinical Rationale: a. A PFM crown has a metal core underneath a ceramic coverage b. Titanium is a metal c. Porcelain veneers are made of glass-matrix ceramics d. Leucite-reinforced ceramics is a glass-matrix ceramic e. PICN means polymer-infiltrated ceramic network. Contains porous feldspathic ceramic infiltrated with a polymer

Question 78

Which of the following is FALSE for feldspathic ceramics?

a. Available in CAD/CAM prefabricated blocks
b. General category of ceramic
c. Feldspathic ceramics are stronger than zirconia
d. Based on silica, alumina, and potassium oxide
e. Used for ceramic veneers

Correct Answer: A — Available in CAD/CAM prefabricated blocks

Clinical Rationale: a. It is still available as CAD-CAM prefabricated blocks b. Felspathic ceramics is included in the general category of glass-matrix ceramics c. Flexural strength of feldspathic ceramics is 80-130 MPa, whereas the flexural strength of zirconia is close to 1,000 MPa d. Its composition is based on silica, alumina, and potassium oxide e. Both feldspathic and lithium disilicate-reinforced ceramics are currently used for ceramic veneers

Question 79

What is the recommended film thickness for dental cements for luting indirect restorations?

a. 1 to 5 µm
b. 5 to 10 µm
c. Depends on the type of cement and the thickness of the indirect restoration
d. ≤50 µm
e. <100 µm

Correct Answer: A — 1 to 5 µm

Clinical Rationale: a. 1–5 µm is very thin and typical only of dental adhesives b. 5–10 µm is too small to allow indirect restorations to fit the respective tooth preparation c. The recommended film thickness for dental cements does not depend on the cement or the restoration d. ≤ 50 µm is the ideal film thickness for dental cement e. <100 µm is too wide and may form a gap between the indirect restoration and the respective tooth preparation

Question 80

Which crown-and-bridge cement requires a separate dental adhesive?

a. Glass ionomer cement
b. Conventional resin cement
c. Self-adhesive resin cement
d. Zinc polycarboxylate cement
e. RMGI

Correct Answer: A — Glass ionomer cement

Clinical Rationale: a. Glass ionomer cement bonds chemically to tooth structure without the need of a dental adhesive b. Conventional resin cement is a composite resin that requires the use of a separate dental adhesive c. Self-adhesive resin cement interacts with the tooth structure without the need of any adhesive d. Zinc polycarboxylate cement bonds chemically to tooth structure without the need of a dental adhesive e. RMGI bonds chemically to tooth structure without the need of a dental adhesive

Question 81

What is the primary monomer involved in denture base fabrication?

a. HEMA
b. BisGMA
c. UDMA
d. TEGDMA
e. MMA

Correct Answer: A — HEMA

Clinical Rationale: a. HEMA is used in dental adhesion b. BisGMA is major monomer used in the composition of composite resins c. UDMA is a monomer mostly used in the composition of composite resins d. TEGDMA is diluent monomer to reduce viscosity used in the composition of composite resins and some dental adhesives e. MMA (methyl methacrylate) is the major monomer in denture-base resin

Question 82

Why are acrylic resin teeth popular for denture fabrication?

a. Can be bonded to denture base
b. Very hard
c. Excellent wear resistance
d. Made of composite resin
e. Poor esthetics

Correct Answer: A — Can be bonded to denture base

Clinical Rationale: a. Acrylic resin teeth can be bonded to the denture base as they have the same chemical makeup as that of the base b. Acrylic resin teeth are soft c. Acrylic resin teeth have poor wear resistance d. Acrylic resin teeth are made of PMMA e. Acrylic resin teeth have good esthetics

Question 83

What is the major problem for some tissue conditioners during the first few days?

a. Bad taste
b. Loss of bonding
c. Discoloration
d. High initial hardness
e. Fungal growth

Correct Answer: A — Bad taste

Clinical Rationale: a. There is no taste produced by the tissue conditioners b. Tissue conditioners are bonded well enough to survive the first few days c. While discoloration can occur, it is not common over the first few days d. Tissue conditioners are softened by plasticizers that rapidly desorb from the conditioner over several days and cause them to become harder e. Although fungal growth can occur along the surfaces of dentures and tissue conditioners, it should not significantly alter the properties of the conditioner

Question 84

How should mouth protectors be cleaned?

a. Cool soap-and-water solution
b. Dilute chlorhexidine solution
c. Denture-cleaning solution
d. Dilute chlorine bleach in water
e. Ultrasonic agitation in baking soda solution

Correct Answer: A — Cool soap-and-water solution

Clinical Rationale: a. Cool soap-and-water solutions are the easiest and cheapest method to cleanse a mouth protector b. Dilute chlorhexidine might stain or otherwise deteriorate the polymer used for the mouth protector c. Denture-cleaning solutions generally are too extreme for the softer polymer used in mouth protectors d. Dilute chlorine bleach in water may attack the polymer in the mouth protector e. Gentle ultrasonic agitation could be used, but there is no benefit to including baking soda solutions for cleaning

Question 85

Mouth protectors are typically fabricated from what material?

a. Impression material
b. Thermoplastic polymer
c. Gutta percha
d. Composite
e. Denture base resin

Correct Answer: A — Impression material

Clinical Rationale: a. Impression material is not used and would be too weak to resist clenching b. Thermoplastic polymer is commonly used to fabricate a mouth protector c. Gutta percha would not be sufficiently flexible d. Composite would not be sufficiently flexible e. Denture base resin would not be sufficiently flexible

Question 86

What does the “A” represent in CAD/CAM?

a. Augmented (or additional)
b. Automated (or automatic)
c. Assisted (or aided)
d. Aged
e. Aligned

Correct Answer: A — Augmented (or additional)

Clinical Rationale: a. Augmented does not make sense b. Automated is not a synonym for “aided or assisted” c. CAD/CAM is the acronym for computer “aided or assisted” design and machining d. Aged does not make sense e. Aligned does not make sense

Question 87

Which process does not harm the protective titanium dioxide surface on dental implants?

a. Air abrasion
b. Abrasive dentifrices
c. Scaling with metal instruments
d. Dental floss
e. APF application

Correct Answer: A — Air abrasion

Clinical Rationale: a. Polishing agents erode the titanium oxide surface of dental implants b. Abrasive dentifrices erode the titanium dioxide surface of dental implants c. Scaling scratches the titanium dioxide surface of dental implants d. Dental floss does not affect the integrity of the titanium dioxide surface of dental implants e. APF attacks the titanium dioxide surface of dental implants

Question 88

Which material is now being used for dental implants in addition to titanium?

a. Alumina
b. Aluminosilicate
c. Silicon carbide
d. Zirconia
e. Porcelain

Correct Answer: E — Porcelain

Clinical Rationale: a. Alumina is not sufficiently strong b. Aluminosilicate is not sufficiently strong c. Silicon carbide strong but is dark and is has poor machinability d. Zirconia is strong and biocompatible to be used for dental implants e. Felspathic ceramic is very weak compared to zirconia and titanium a. Inkjet printers can be used to build up (additive) restorations in three dimensions b. CAD/CAM machining of composite blocks is a subtractive process because material is being removed c. CAD/CAM machining of zirconia blocks is a subtractive process because material is being removed d. Casting does not involve milling and is thus not categorized as additive or subtractive e. ‘a’ is the correct answer

Question 89

Which dental fabrication process is classified as additive?

a. Inkjet printing
b. CAD/CAM machining of composite blocks
c. CAD/CAM machining of zirconia blocks
d. Casting
e. None of the above

Correct Answer: A — Inkjet printing

Clinical Rationale: a. MMA (liquid) is mixed with PMMA (powder) b. Allergic reactions to unreacted residual monomer or benzoic acid have been reported in newly processed dentures c. Faily high volumetric polymerization shrinkage—5% to 7% d. Poor extortion resistance at higher temperatures e. Poor scratch resistance

Question 90

Which statement applies to acrylic denture bases?

a. Contain HEMA and PMMA
b. Unreacted monomer may cause allergic reactions
c. Very low polymerization contraction
d. Resist distortion at higher temperatures
e. Excellent scratch resistance

Correct Answer: A — Contain HEMA and PMMA

Clinical Rationale: a. Composite restorations are strong but may chip over time and can be easily repaired b. All ceramic restorations have good abrasion resistance c. Veneers must be bonded to the tooth structure. They require enamel etching and dental adhesive for bonding to tooth structure d. Clinical outcomes with veneers are very esthetic e. All the above

Question 91

Which of the following applies to veneers?

a. Composite veneers may chip over time
b. Ceramic veneers have good abrasion resistance
c. Veneers require enamel etching and dental adhesive for bonding to tooth structure
d. Veneers result in good esthetic outcomes
e. All of the above apply to veneers

Correct Answer: A — Composite veneers may chip over time

Clinical Rationale: a. Fiber-reinforced resin blocks are available for CAD-CAM milling of root canal posts b. Resin cements for bridges must be mixed prior to use c. PFM crowns are made of two different materials that are bonded to each other d. Custom-made bleaching trays are made of very thin and flexible ethylene-vinyl acetate. They must be fabricated manually e. RMGI must be mixed prior to use

Question 92

Which material is available in blocks for CAD/CAM milling?

a. Fiber-reinforced resin for root canal posts
b. Resin cements for bridges
c. PFM crown
d. Custom-made bleaching tray
e. RMGI

Correct Answer: A — Fiber-reinforced resin for root canal posts

Clinical Rationale: a. Sandblasting does not improve the fit of the crown b. Sandblasting should be done after try-in of the crown to avoid contamination of the intaglio surface with saliva c. Sandblasting is not used to clean crowns d. Sandblasting adds micromechanical retention to the intaglio surface of the restoration e. Sandblasting does not improve the final color

Question 93

Why are zirconia crowns sandblasted prior to cementation?

a. To obtain a better fit of the crown
b. To help with the try-in of the crown
c. To clean the crown
d. To add micromechanical retention to the crown
e. To improve the final color

Correct Answer: A — To obtain a better fit of the crown

Clinical Rationale: a. Etching with hydrofluoric acid does not improve the stability of the surface b. Etching the intaglio surface of glass-matrix crowns with hydrofluoric acid adds micromechanical retention c. Etching with hydrofluoric acid does not result in better fit of the crown d. Etching with hydrofluoric acid does not help with the try-in of the crown e. Etching with hydrofluoric acid is not used to clean the crown

Question 94

Why are glass-matrix crowns etched with hydrofluoric acid prior to cementation?

a. To improve the stability of the surface
b. To add micromechanical retention to the crown
c. To obtain a better fit of the crown
d. To help with the try-in of the crown
e. To clean the crown

Correct Answer: A — To improve the stability of the surface

Clinical Rationale: a. Protection of hard and soft tissues is a physiological objective of provisional materials b. Stabilization of tooth is a physiological objective of provisional materials c. Patient comfort is a physiological objective of provisional materials d. Pulpal protection is a physiological objective of provisional materials e. All options are correct

Question 95

Which are the physiological objectives of provisional materials?

a. Protection of hard and soft tissues
b. Stabilization of tooth
c. Patient comfort
d. Pulpal protection
e. All of the above

Correct Answer: A — Protection of hard and soft tissues

Clinical Rationale: a. Silane does not work as polymerization initiator b. Silane does not work as polymerization accelerator c. Silane provides chemical bonding between the surface of the filler particles and the resin in the matrix of a composite resin d. Dental adhesives bond the polymer matrix to the tooth structure. e. TEGDMA serves as lower-molecular-weight diluent

Question 96

What is the role of the silane in the composition of a composite resin?

a. Polymerization initiator
b. Polymerization accelerator
c. Bonds chemically the filler particles to the polymer matrix
d. Bonds chemically the polymer matrix to the tooth structure
e. Serves as lower-molecular-weight diluent

Correct Answer: A — Polymerization initiator

Clinical Rationale: a. Biocompatibility the ability of a biomaterial to perform its desired function without eliciting any undesirable local or systemic effects b. Infection is the invasion and growth of microorganisms in the body that cause harm c. Allergy is an immune response to a substance (known as allergen) present in the environment that are usually harmless to others (like allergy to methacrylates present in resin composites) d. Toxicity is the ability of a substance to cause harmful health effects on single cell, a group of cells, or an organ system e. Rebuilding tissues in the laboratory to be implanted

Question 97

What is an immune response to a substance present in the environment that is usually harmless to others?

a. Biocompatibility
b. Infection
c. Allergy
d. Toxicity
e. Tissue-engineered replacement

Correct Answer: A — Biocompatibility

Clinical Rationale: a. Cells are also required b. For creation of new tissue (tissue engineering) there must be cells, signals, and scaffolds c. Signals are also required d. Signals are also required e. Signals are also required

Question 98

What are the major requirements for tissue engineering?

a. Collagen, hydroxyapatite, and signals
b. Cells, signals, and scaffolds
c. Cells and collagen
d. Cells and bone
e. Cells and scaffolds

Correct Answer: A — Collagen, hydroxyapatite, and signals

Clinical Rationale: a. Xenograft is tissue replacement that comes from different species b. Synthetic tissue is made from nonliving materials and remains artificial c. Tissue engineering replacement is produced in the laboratory or built in situ from cells, signals, and scaffold material d. Autograft is tissue replacement that comes from one’s own body e. Allograft is tissue replacement that comes from the same species, usually cadavers

Question 99

Which biomaterial utilizes tissue from a cadaver?

a. Xenograft
b. Synthetic tissue
c. Engineered tissue
d. Autograft
e. Allograft

Correct Answer: E — Allograft

Clinical Rationale: a. Xenograft is tissue replacement that comes from different species b. Synthetic tissue is made from nonliving materials and remains artificial c. Tissue engineering replacement is produced in the laboratory or built in situ from cells, signals, and scaffold material d. Autograft is tissue replacement that comes from one’s own body e. Allograft is tissue replacement that comes from the same species, usually cadavers

Question 100

Which of the following is not a scaffold material for tissue engineering?

a. Gold wire mesh
b. Polylactic acid (PLA)
c. Bioglass
d. Collagen
e. Bone chips

Correct Answer: A — Gold wire mesh

Clinical Rationale: a. Gold mesh is not used as a scaffold b. Polylactic acid (PLA) scaffolds are common for tissue engineering c. Bioglass is a bioactive ceramic material that is commonly used to help form new bone d. Collagen is a common scaffold for both soft- and hard-tissue engineering e. Bone chips have been used as a scaffold for forming new bone